HIM superuser training

Tuesday, August 3, 2021 5:36 PM

close main valve CA cylinder
release inter valve

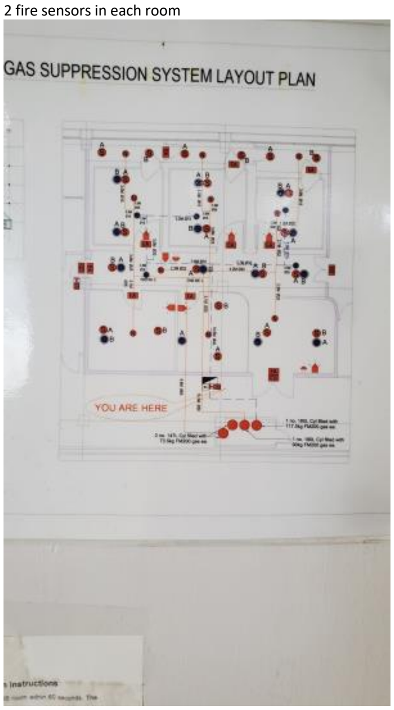
to check leakage, open main valve then shut, wait 2min pressure stays

1. Alarms
   a. O2 monitor
   Outside HIM room

   Channel = 1, monitor reading 20~21%
   In case of N2 leakage, O2 level can drop below 20%. Evacuate personnel

   B. Fire alarm in corridor.
   2 fire sensors in each room

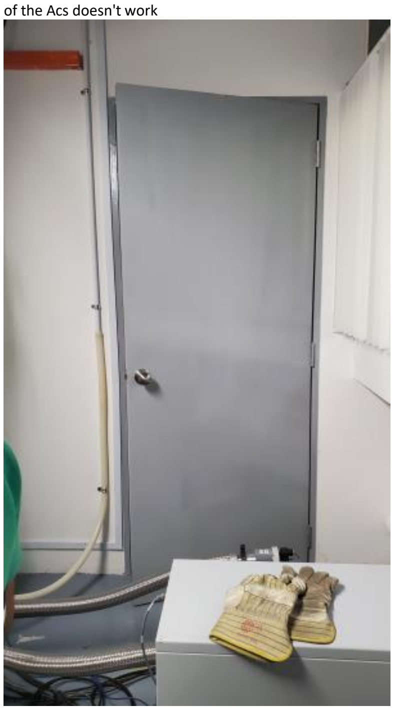
   In case of fire, evacuate to fire assembly point. When fire alarm is triggered, gas will release in 60s to put out fire. Near fire alarm in corridor, there are 2 switches: abort and release. If confirmed no fire or smoke, press abort within 60s to reset fire alarm.

2. Backroom
   Some cylinders and electronics are stored in backroom.
   Double doors to backroom for soundproof
   Big cylinder is for condensed air to control column valves. Small cylinder is for dry N2 to purge loading

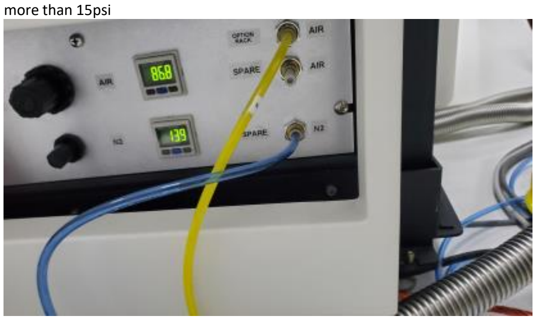
When disconnecting the cylinder, close dewar => table suspension to prevent sudden drop in pressure

2. Backroom
   Some cylinders and electronics are stored in backroom.
   Double doors to backroom for soundproof
   Big cylinder is for condensed air to control column valves. Small cylinder is for dry N2 to purge loading chamber

   Make sure AC is on lowest setting all time. Make sure the door connecting next room is open in case any one of the Acs doesn't work



   Readings for condensed air and N2 are at back of HIM. Air should be 80~90psi, N2 should be 9~13psi, no more than 15psi



   To change cylinder:

When disconnecting the cylinder, close dewar => table suspension to prevent sudden drop in pressure

When the pressure reading is <100psi, or the main chamber CA reading behind the machine is <80 (it will turn red), we need to change cylinder.

i. Decrease valve 1 to loose to disconnect main chamber with the cylinder
ii. Shut valve 2 to fully closed
iii. Use spanner to loose 3. disconnect the cylinder completely
iv. Clean the teflon tape around the connection
v. Prepare new cylinder
vi. Wrap new teflon tape around the connection: reverse direction towards tightening
vii. Open main valve 2 to see pressure reading increase. Then shut main valve 2. wait for 1min, if the pressure reading is stable, it means no leakage
viii. Open main valve 2 fully. Increase valve 1 to ~95psi. Check the main chamber reading to make sure it's around 86

- Check the pressure in next 3 days. If the pressure is dropping too quickly then check leakage again.
- Zen software may report error if the air pressure is too low. Restart the system, check hardware communication, then check trimer
- Trimer extraction voltage -33keV. Always check trimer on conducting sample (standard Au mark 2)

3. Liquid N2 refill

Auto refill to HIM is set twice a day from the big tank. 1 tank of LN2 lasts ~16d. To refill LN2 tank:

i. In ZEN software, LN2 tab -> begin servicing to override auto refill
ii. Disconnect the tank. After moving out the tank, HIM can be used during refill. Moving tank in and out will disturb the beam. Stand-by the GFIS column when moving the tank.
iii. After vendors refill the tank, sign DO and pass to Teo. Make sure the valves are up

Press done servicing to activate auto refill again.

4. Operation

i. Check lens2 power before loading samples
ii. Always stand-by the column power when not using

When you book a slot for training and Trimer formation, please type "For Training : Ms XX email and Mr XY Email" and "For Maintenance" , respectively. These words will let Shou Xuan know that these booked slots will not be billed under your name.